

Conjugacy Exploration — a picture book

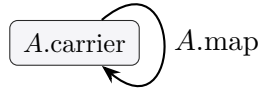
Frobenius shift, intertwiners, and the Gold / Kasami test

Visual summary of `Dobbertin1999MVP/ShiftConjugacy.lean`

One line. Kasami and Gold are *not* the same map up to a Frobenius shift. The real bridge is a bigger kind of map (an *intertwiner*). Every claim below is machine-checked in Lean.

1. The players: a set with a self-map

An **object** is a set with one arrow going back into itself.



A **morphism** $L : A \rightarrow B$ is a map that *respects* the self-maps. The square commutes: go right-then-down = go down-then-right.

$$\begin{array}{ccc} A.\text{carrier} & \xrightarrow{L} & B.\text{carrier} \\ A.\text{map} \downarrow & & \downarrow B.\text{map} \\ A.\text{carrier} & \xrightarrow{L} & B.\text{carrier} \end{array}$$

Lean: `structure Endo`, `structure EndoHom` with law $B.\text{map}(L x) = L(A.\text{map } x)$.
These objects and arrows form a category (`instance : Category Endo`).

2. Conjugacy = isomorphism

If L is a **bijection** then the arrow flips for free: L^{-1} is again a morphism. So a two-way arrow means the two self-maps are **conjugate**.

$$\begin{array}{ccc} A.\text{carrier} & \xrightleftharpoons[L^{-1}]{L} & B.\text{carrier} \\ A.\text{map} \downarrow & & \downarrow B.\text{map} \\ A.\text{carrier} & \xrightleftharpoons[L^{-1}]{L} & B.\text{carrier} \end{array} \iff B.\text{map} = L \circ A.\text{map} \circ L^{-1}$$

Lean: `isoOfEquiv` (bijective intertwiner $\Rightarrow$ iso), `Conjugate`.
The paper's substitution ψ_β (MCM $\Leftrightarrow$ Kasami) is one such isomorphism.

3. Frobenius = the shift on exponents

Take power maps $x \mapsto x^e$. On the *exponent* e , squaring ($x \mapsto x^2$) is just **doubling** $e \mapsto 2e$.

$$\begin{array}{ccc} x & \xrightarrow{(\cdot)^e} & x^e \\ & \searrow (\cdot)^{2e} & \downarrow (\cdot)^2 \\ & & x^{2e} \end{array} \quad x^{2e} = (x^e)^2$$

Every Frobenius power $x \mapsto x^{2^i}$ is a self-arrow of the object $x \mapsto x^e$ (it slides past the power map):

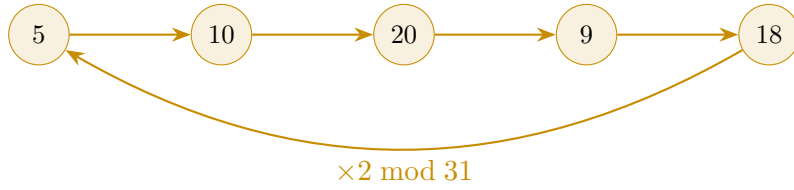
$$\begin{array}{ccc}
 x & \xrightarrow{(\cdot)^{2^i}} & x^{2^i} \\
 (\cdot)^e \downarrow & & \downarrow (\cdot)^e \\
 x^e & \xrightarrow{(\cdot)^{2^i}} & (x^e)^{2^i} = (x^{2^i})^e
 \end{array}$$

Lean: `frobenius_realizes_shift`, `frob_pow_intertwines`, `frobPowHom`.

4. Orbits of doubling = cyclotomic cosets

Work mod $m = 2^n - 1$. Doubling $e \mapsto 2e$ marches an exponent around a cycle. Running $i < n$ gives *all* n field automorphisms of $\mathbb{F}_{2^n}$. Each cycle is a **cyclotomic coset**.

Example $n = 5$, $m = 31$. The Gold exponent 5 walks a 5-cycle:



5. The test: does Gold meet Kasami?

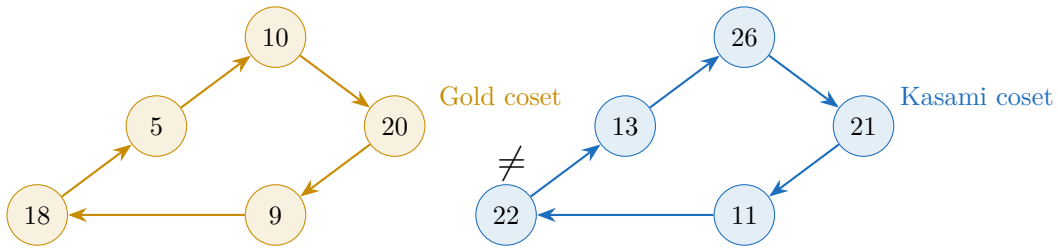
Gold exponent $2^k + 1$ vs Kasami exponent $2^{2k} - 2^k + 1$.

At $k = 1$ they collapse to the same number (3). No genuine Kasami yet.

$$\begin{array}{ccc}
 \textcircled{3} & = & \textcircled{3} \\
 \text{Gold } 2^1+1 & & \text{Kasami } 2^2-2+1
 \end{array}$$

Lean: `kasami_eq_gold_at_one`.

At $k \geq 2$ they live in different cycles — no amount of doubling (no Frobenius power) moves one onto the other. Case $n = 5$, $k = 2$: Gold = 5, Kasami = 13, mod 31.



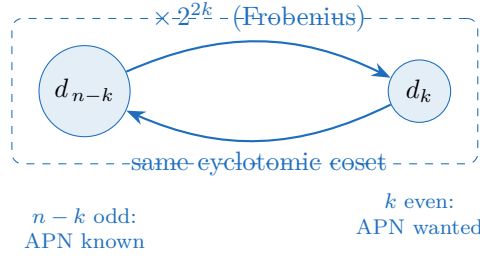
Two disjoint cycles $\Rightarrow$ no Frobenius power links them. Checked for $(n, k) = (5, 2), (7, 2), (7, 3), (11, 2), (13, 3)$: `gold_kasami_not_frobEquiv_*`.

6. Bonus: the *same* shift, put to work (even k)

The conjugacy exploration says Frobenius **cannot** link two *different* cosets (Gold vs Kasami). The flip side: when two exponents sit in the *same* coset, Frobenius links them for free — and that is exactly the trick that removes the “ k odd” restriction from Kasami APN.

Kasami with parameter k and with parameter $n - k$ live in **one** coset:

$$d_k = d_{n-k} \cdot 2^{2k} \pmod{2^n - 1}, \quad \text{so} \quad x^{d_k} = (x^{d_{n-k}})^{2^{2k}} = \text{Frob}_{2k}(x^{d_{n-k}}).$$



Frobenius is an additive bijection, so it **preserves APN**. Transfer from the odd side ($n - k$) to the even side (k). Done.

Lean: `kasami_pow_frob_identity`, `apn_frob_twist`, `kasami_apn_of_complement`, `kasami_is_apn_general` (and `IsAPN.comp_frobenius` in the differential-uniformity library).

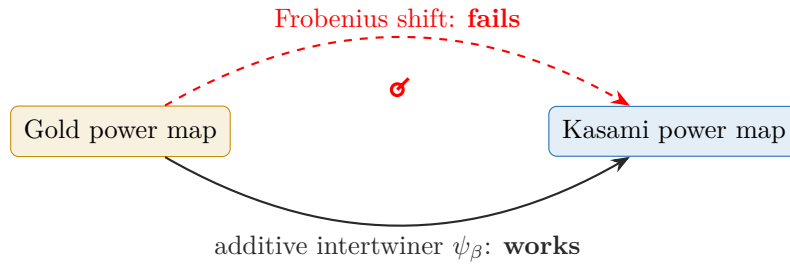
Does Dobbertin use this move? Yes — word for word. Right after Corollary 2 he writes: “we may assume that k is odd without loss of generality. (In fact, replace k by $n - k$ if k is even. This replacement does not change the APN property.)” His justification is *cyclotomic equivalence* — his “cyclic shift” is our Frobenius shift. The same $k \leftrightarrow n - k$ symmetry reappears for his difference sets: “ B does not change if (k, k') is replaced by $(n - k, n - k')$.”

How it relates to the MCM $\Leftrightarrow$ Kasami bridge. Two different uses of *one* idea (conjugacy in the category of self-maps):

- **Inside** a coset ($k \leftrightarrow n - k$): the intertwiner *is* a Frobenius power. Cheap, and it patches the parity gap so the MCM proof (which needs k odd) covers *all* k .
- **Across** cosets (MCM $\Leftrightarrow$ Kasami, ψ_β): the intertwiner is additive but *not* Frobenius — the exploration shows Frobenius alone cannot do it.

So the Frobenius twist is the *complement* of ψ_β , not a rival: it is the special (same-coset) case of the same conjugacy umbrella.

7. Verdict



The shift picture is the right *idea*, but the umbrella map is a genuine morphism of the conjugacy category — an additive (linearized) intertwiner — **not** a power of Frobenius.